

Pose-Based Human Activity Classifier

Computer Vision | Complex Computing Problem (CLO-3) | Rule-based pose pipeline with MediaPipe

99.35%
Accuracy

613
Frames analysed

2
Activities

4
Misclassified

1 Overview

This project detects human poses from a video, analyses joint geometry, and classifies the activity in every frame.

Video source: own pre-recorded clip **activity.mp4** (~20.5 s, 613 frames @ 30 fps) of a single person repeatedly standing up and sitting down. The four-stage pipeline is shown below.



2 Methodology

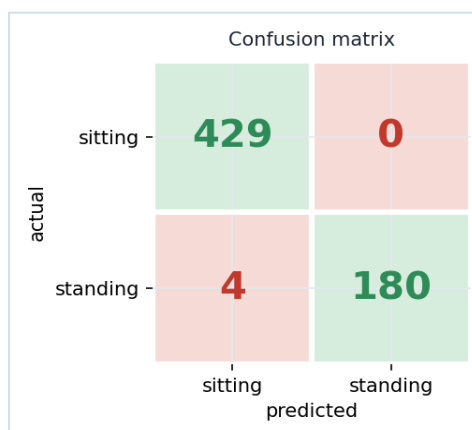
Pose detection & pre-processing. The pre-trained MediaPipe Pose model returns 33 landmarks per frame. Missing detections are linearly interpolated and each landmark trajectory is de-jittered with a Savitzky–Golay filter (window 11, order 3) along the time axis, which acts as a low-pass smoother that preserves real motion.

Joint-angle computation. For three landmarks A–B–C, the angle at B is $\theta = \arccos((\mathbf{BA} \cdot \mathbf{BC}) / (|\mathbf{BA}| |\mathbf{BC}|))$, evaluated in pixel coordinates. Three angles are tracked (left/right averaged): **knee** (hip–knee–ankle), **hip** (shoulder–hip–knee) and **elbow** (shoulder–elbow–wrist).

Rule-based classifier. A frame is labelled **STANDING** when (knee > 130°) AND (hip > 130°), otherwise **SITTING**. Using two joints rather than the knee alone adds robustness to single-joint noise.

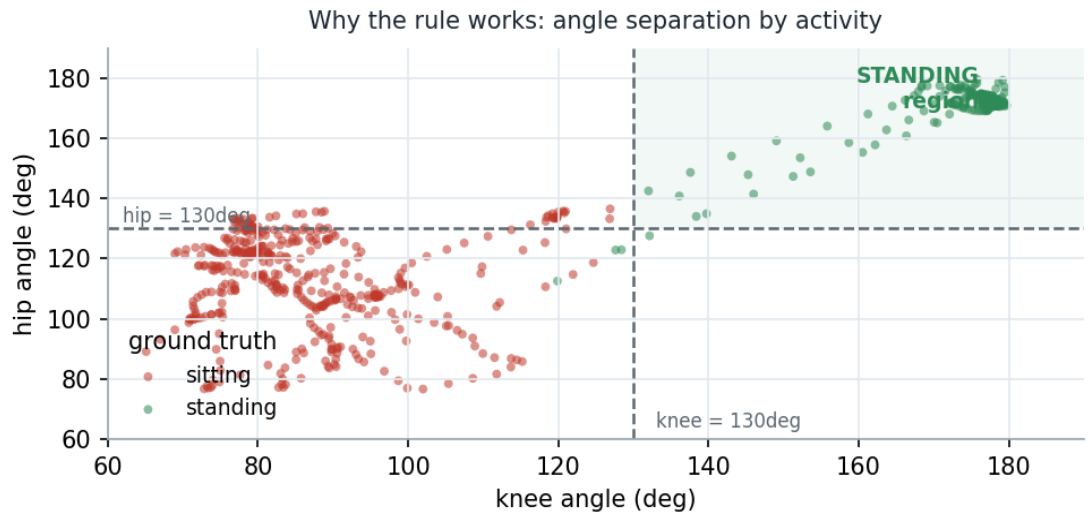
3 Results

The classifier reaches **99.35%** accuracy (609/613 frames) against a manually-labelled per-frame ground truth. The confusion matrix and per-class metrics below show the only errors are 4 transition frames [140, 141, 416, 417].

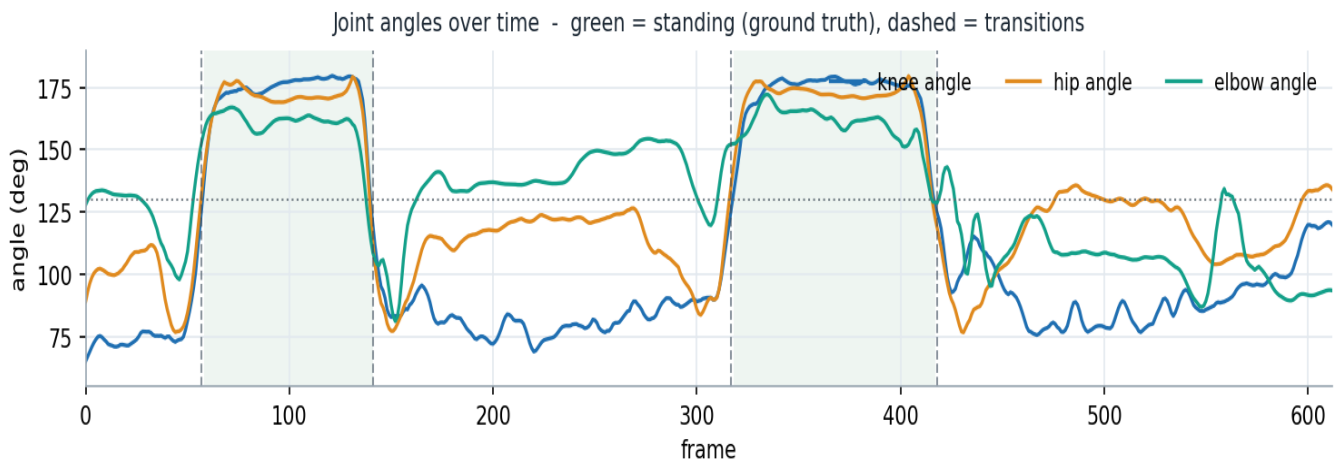


Class	Precision	Recall	F1-score	Support
Sitting	0.991	1.000	0.995	429
Standing	1.000	0.978	0.989	184
Macro avg	0.995	0.989	0.992	613

Activity separation in knee-hip angle space.

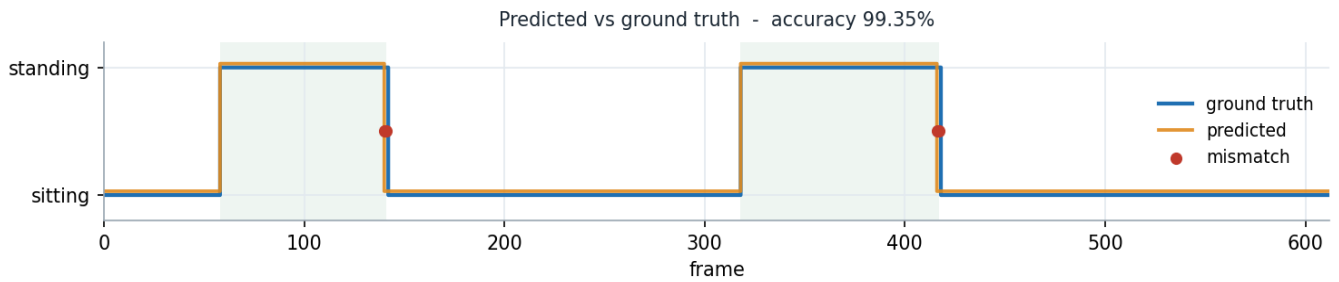


Joint angles tracked across the clip.

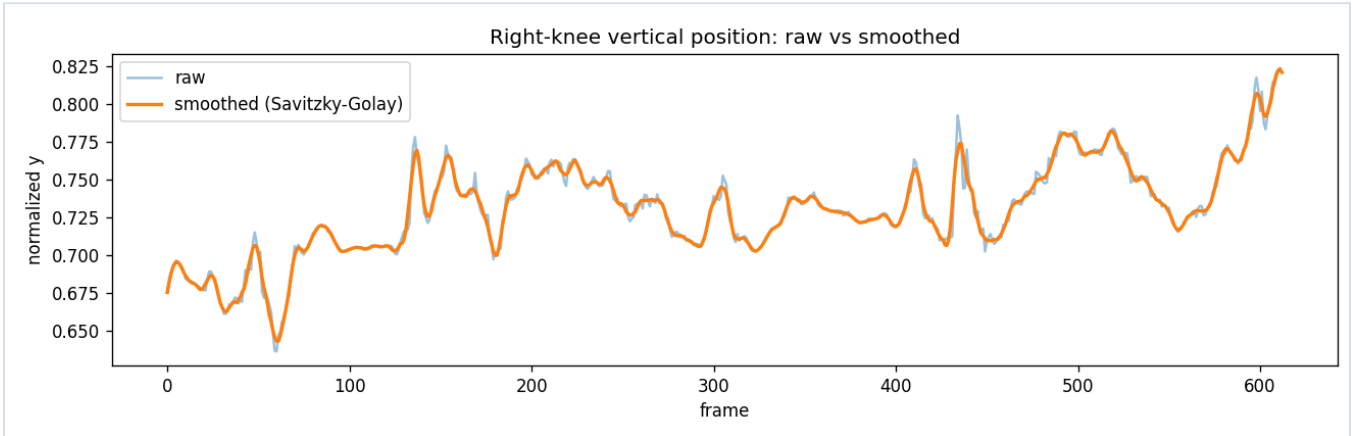


4 Tracking, Transitions & Demo

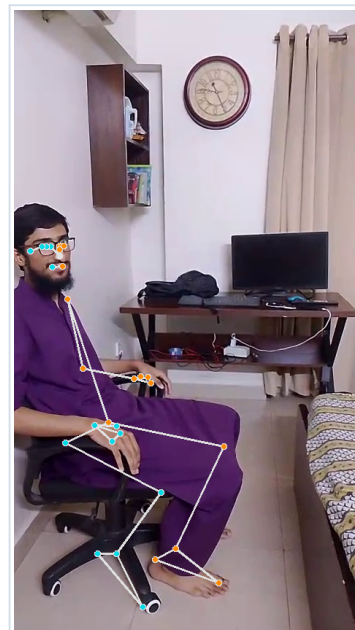
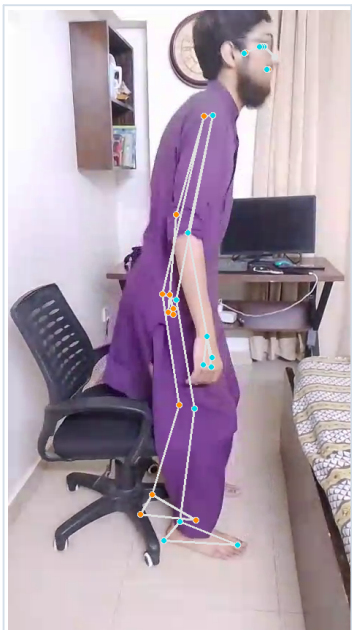
Per-frame prediction vs ground truth.



Smoothing reduces keypoint jitter (right-knee vertical position).



Skeleton overlay produced by the pipeline: standing (left) and sitting (right).



5 Observations

- MediaPipe detected the subject in all 613 frames; Savitzky–Golay smoothing removed jitter without distorting the underlying motion.
- Knee and hip angles separate the two activities cleanly (standing $\sim 150\text{--}180^\circ$, sitting $\sim 65\text{--}120^\circ$); the elbow angle is comparatively uninformative for stand/sit.
- Every classification error falls on a transition frame (mid stand-up / sit-down), where the posture is genuinely ambiguous — exactly where a hard threshold is expected to disagree with a discrete human label.